Fpt University

Introduction to Java Web Application Development (PRJ301)

Kiều Trọng Khánh

https://lms-hcm.fpt.edu.vn/
KhanhKT

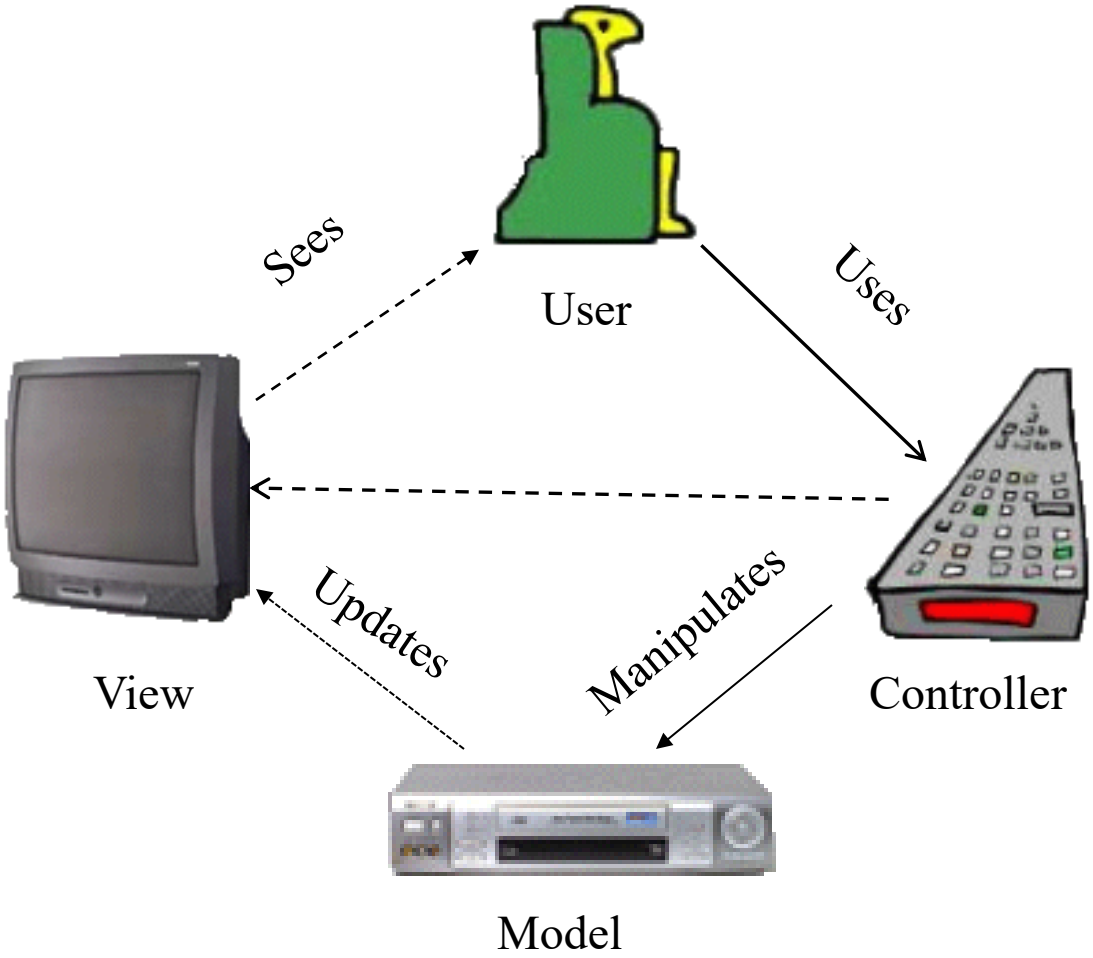
Prerequisites

- **Completed:**
  - DBI202 (Database Systems)
  - PRO192 (Object Oriented Programming)

Course Objectives

MVC Design Pattern

Model – View – Controller



This is a MVC Model

Course Objectives

- The MVC architecture is applied in web application

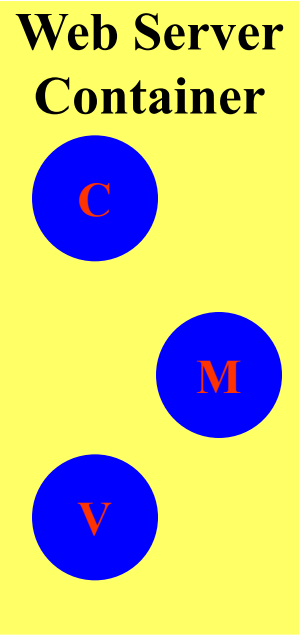
Course Objectives

- The MVC architecture is applied in web application

Web Server Container

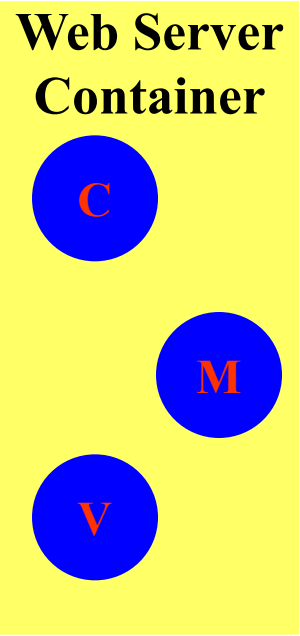
Course Objectives

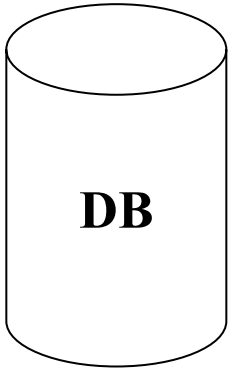
- The MVC architecture is applied in web application



Course Objectives

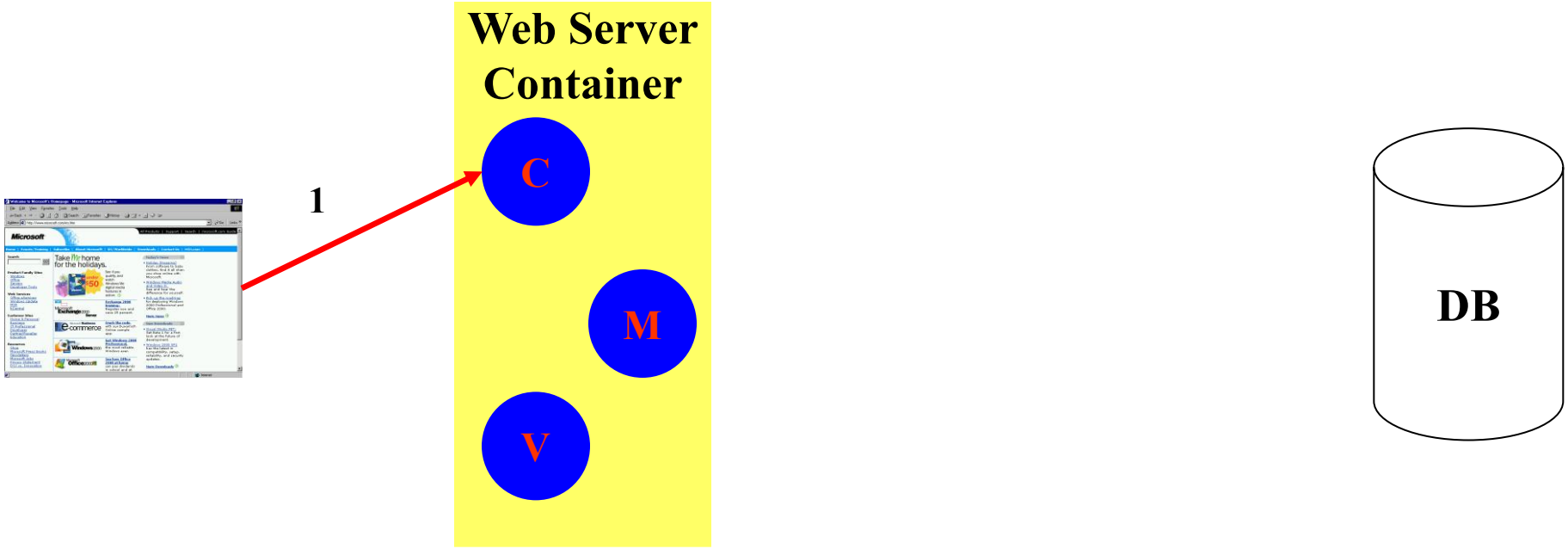
- The MVC architecture is applied in web application





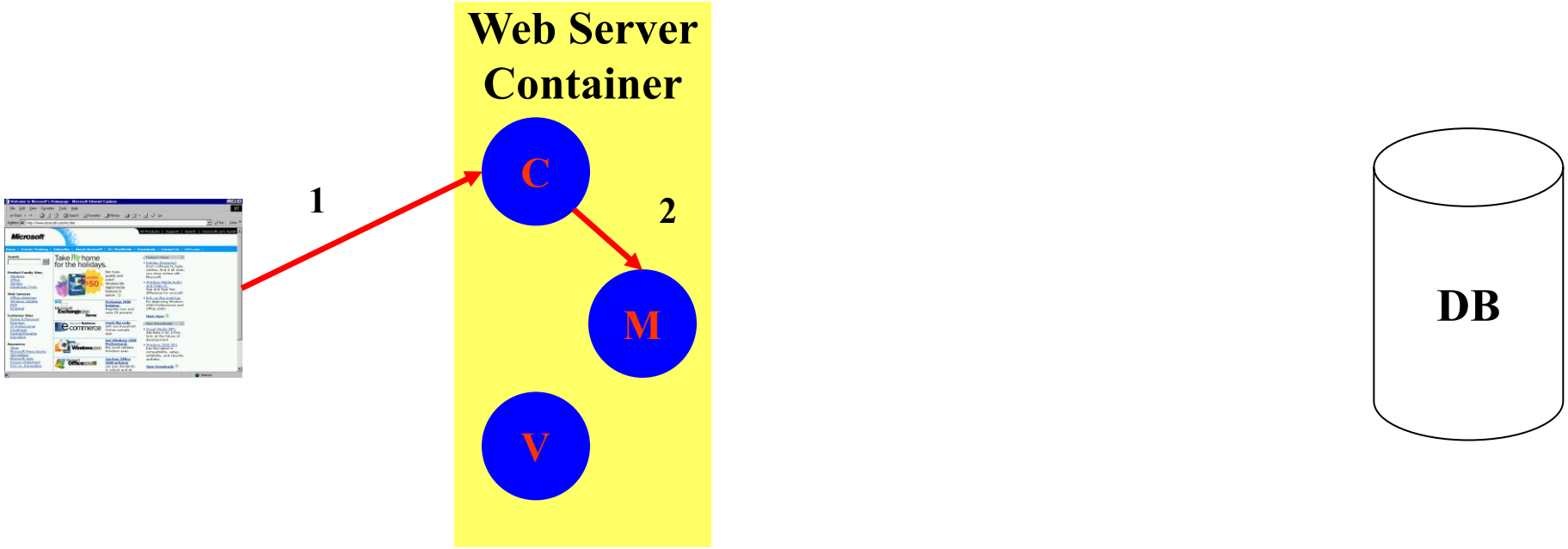
Course Objectives

- The MVC architecture is applied in web application



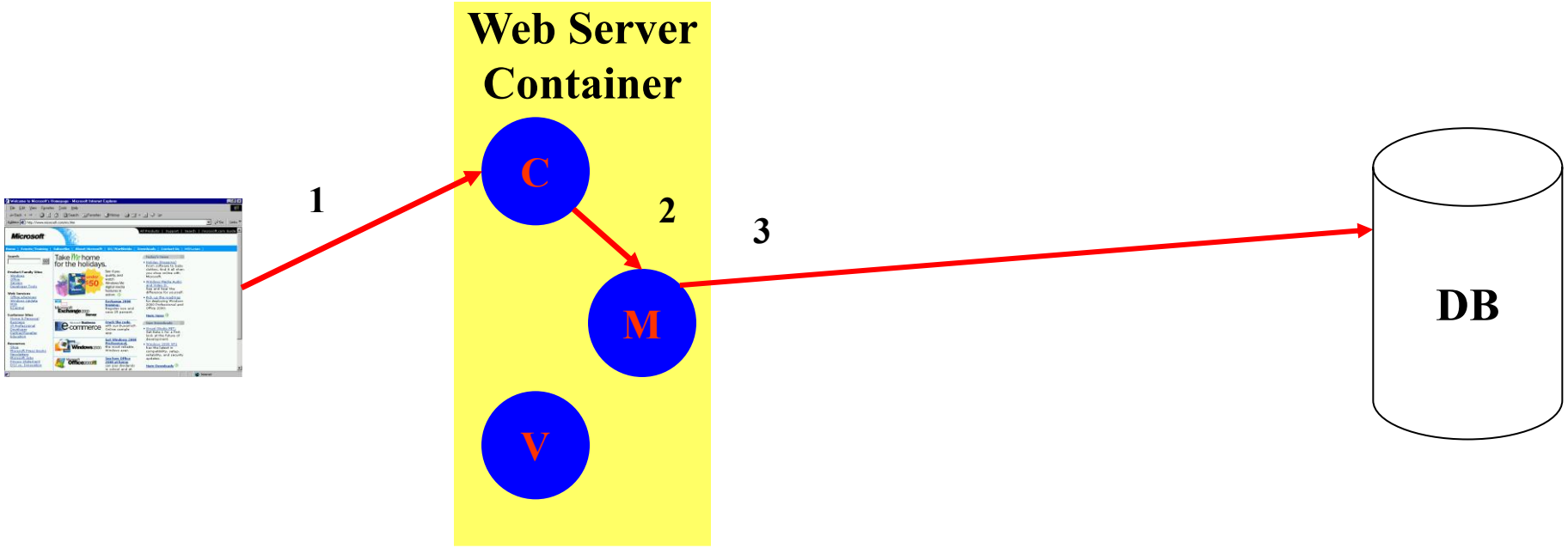
Course Objectives

- The MVC architecture is applied in web application



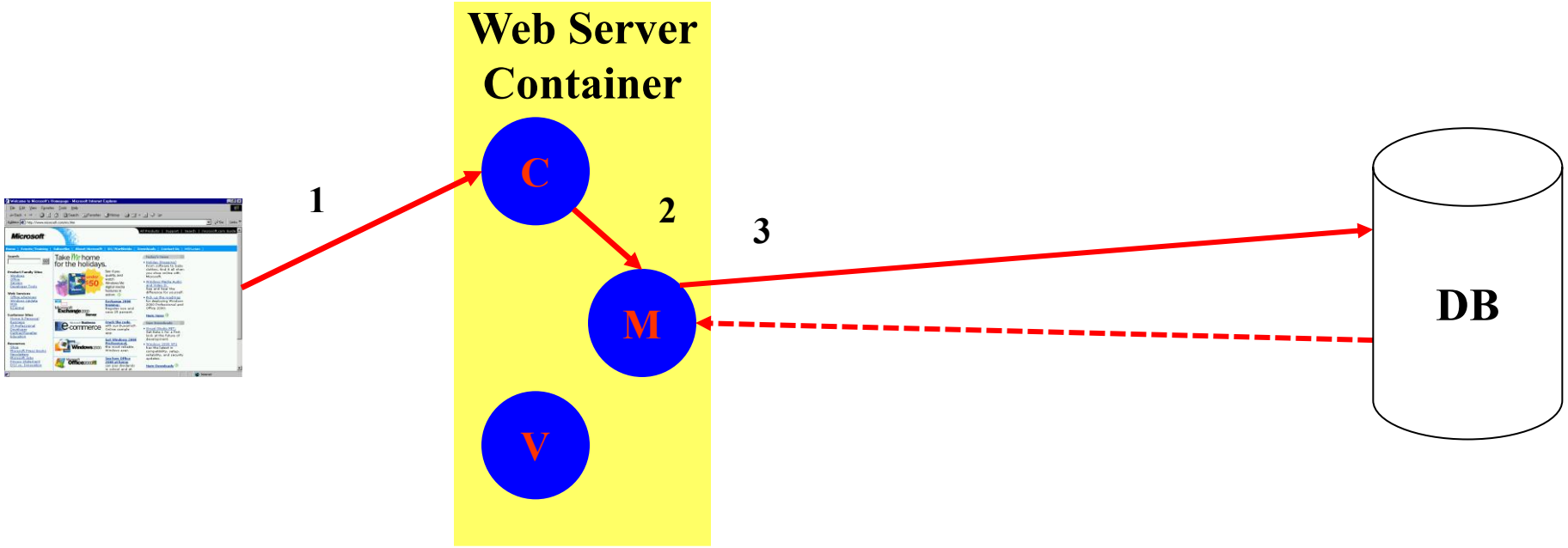
Course Objectives

- The MVC architecture is applied in web application



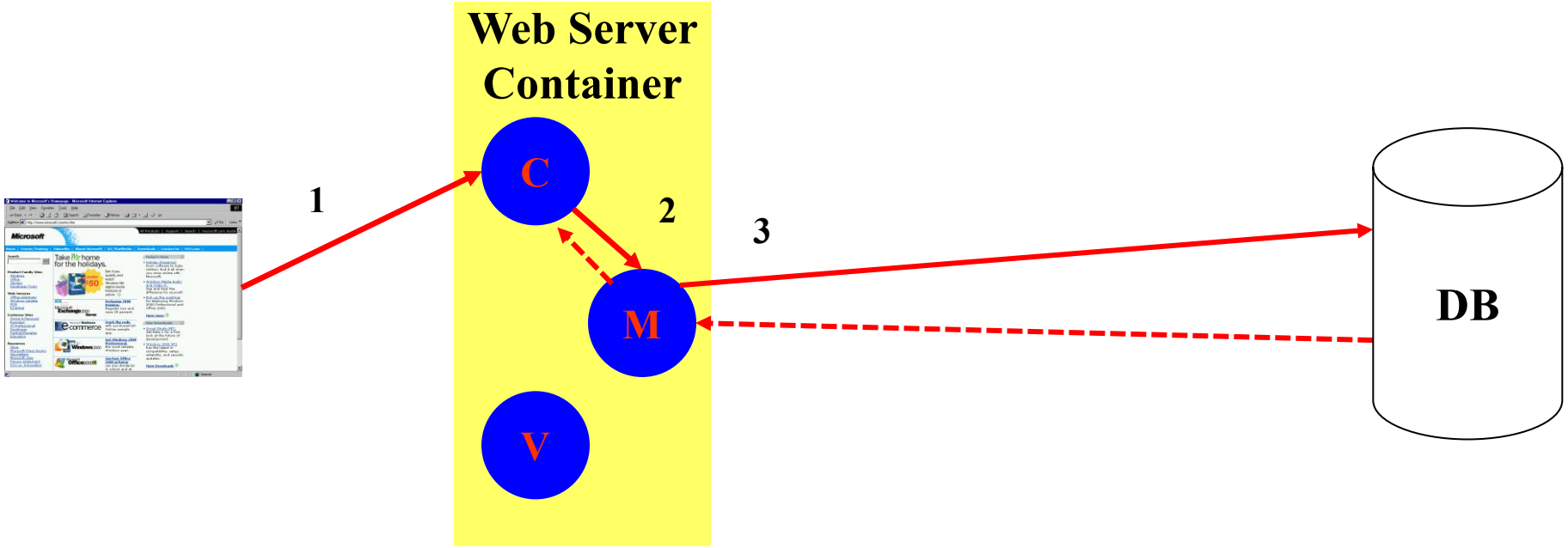
Course Objectives

- The MVC architecture is applied in web application



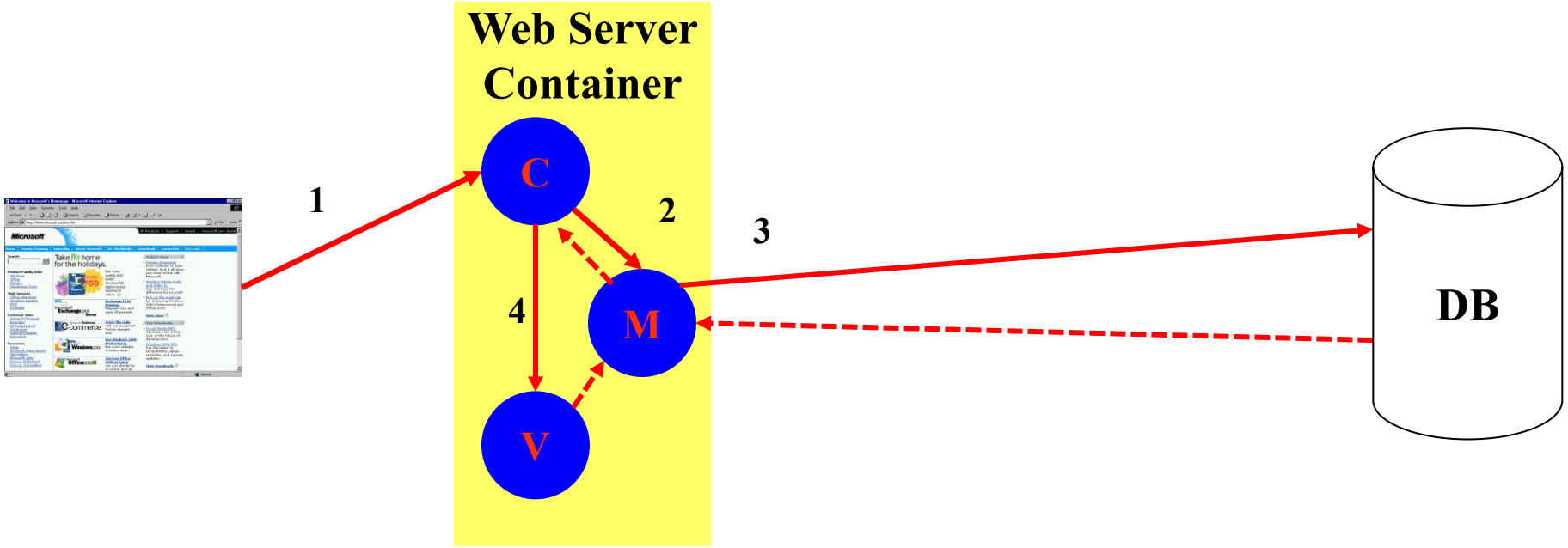
Course Objectives

- The MVC architecture is applied in web application



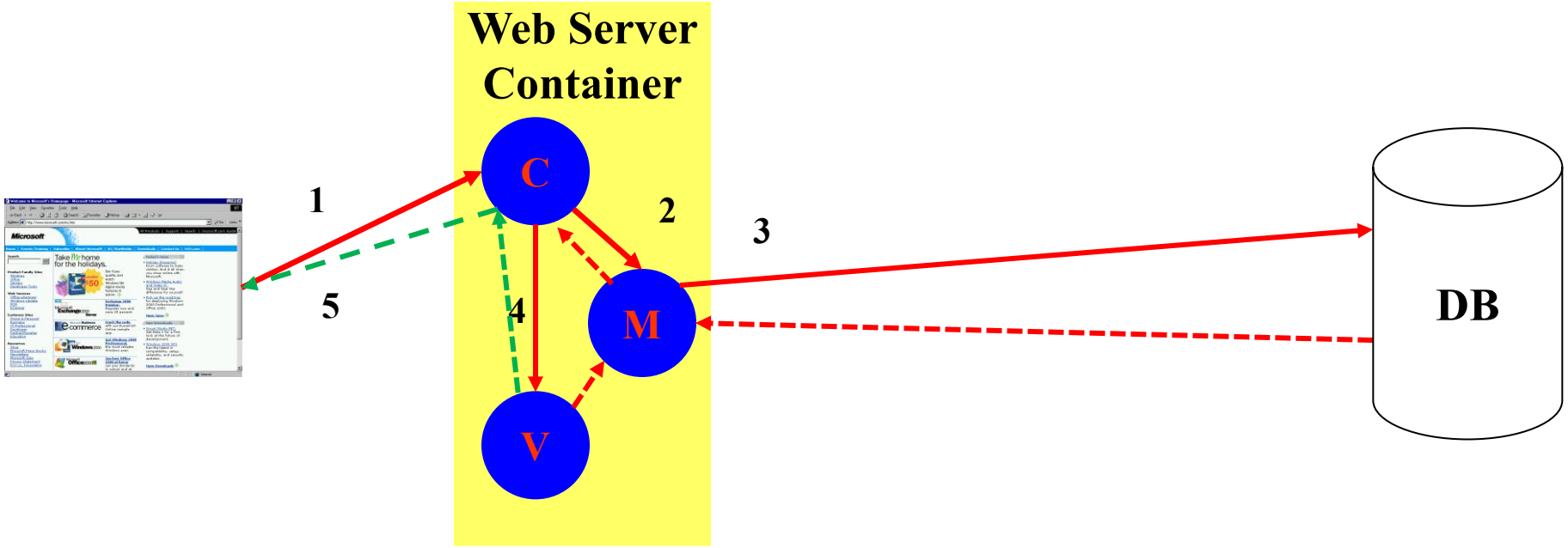
Course Objectives

- The MVC architecture is applied in web application

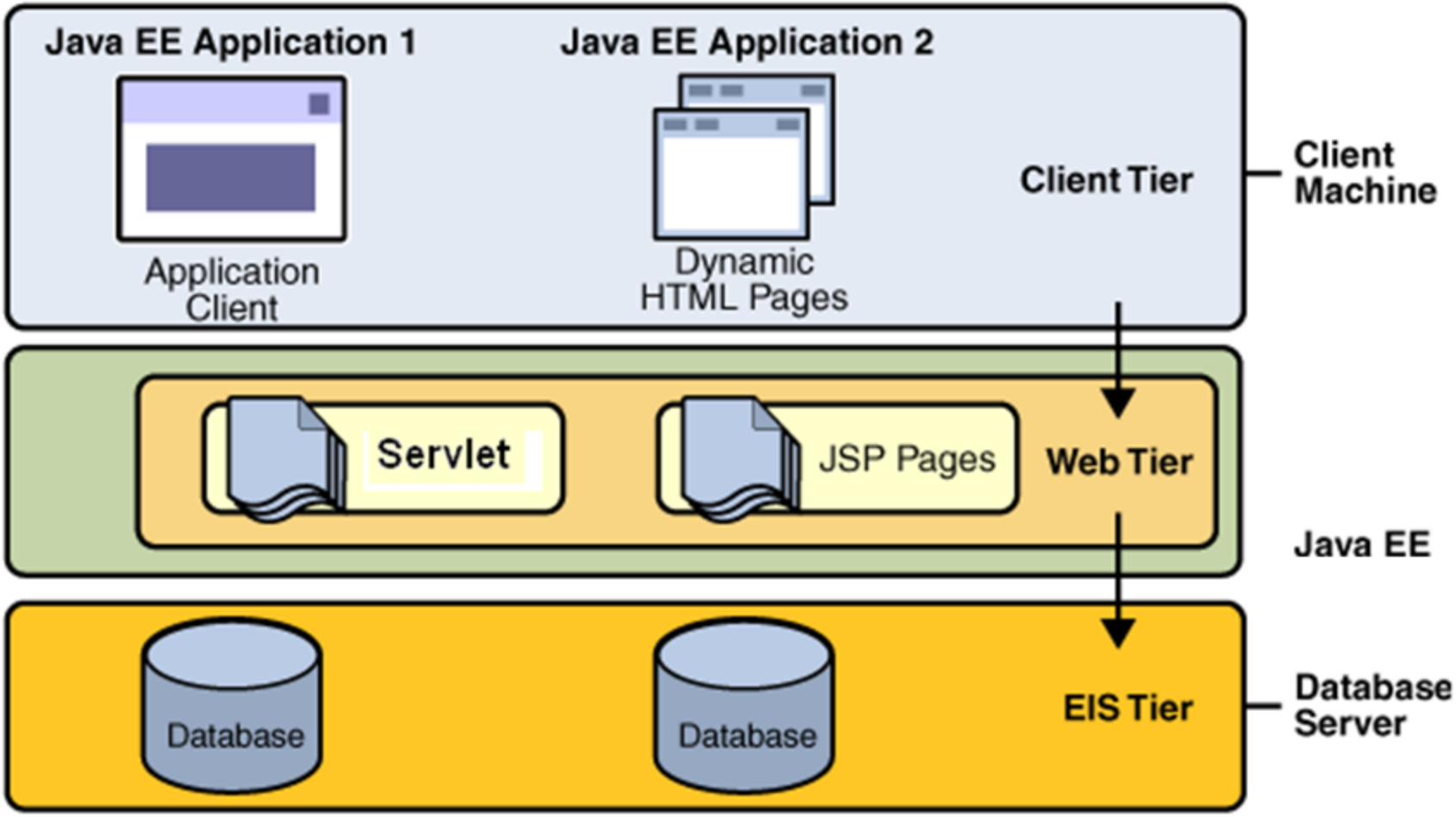


Course Objectives

- The MVC architecture is applied in web application

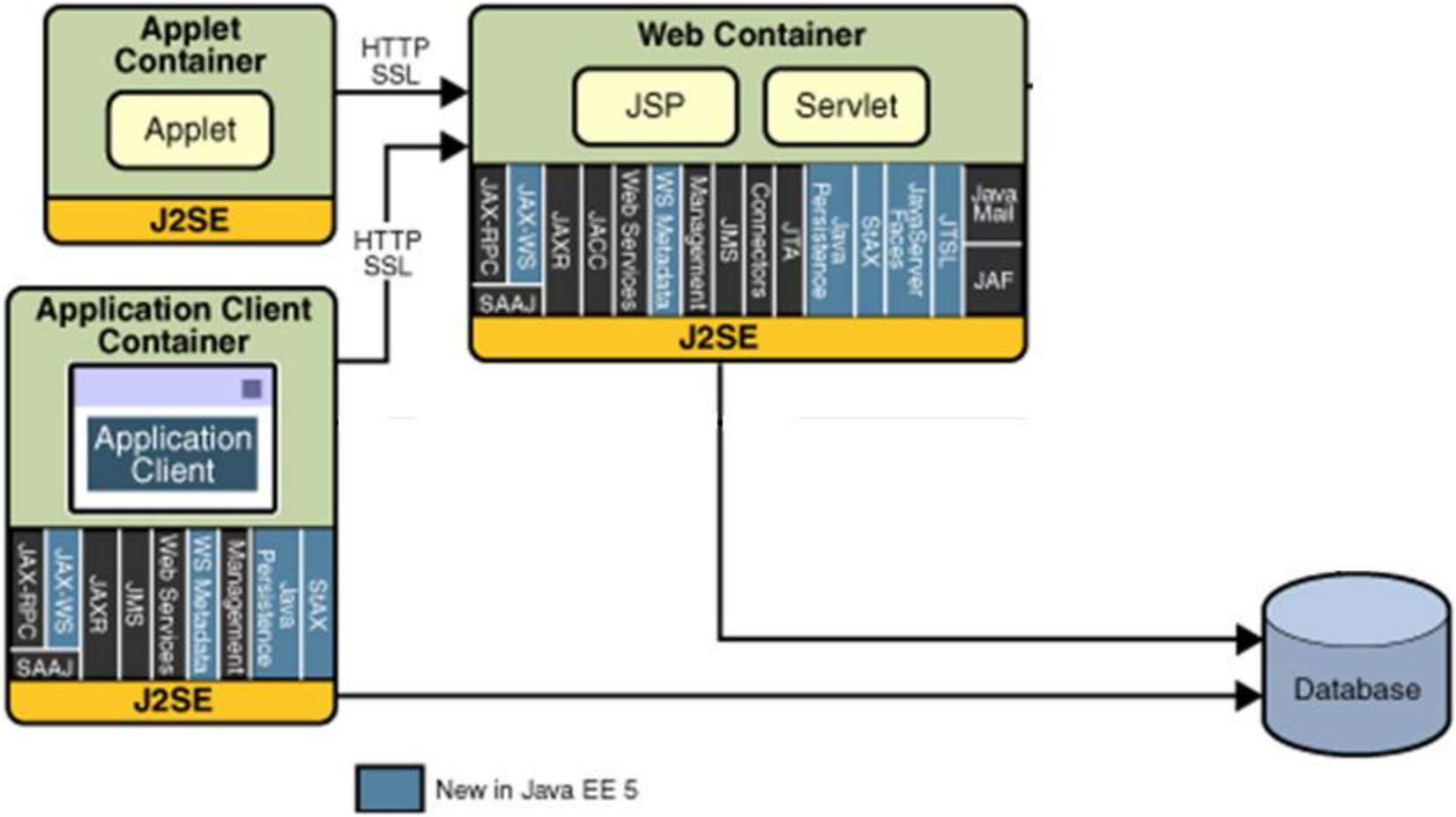


Course Objectives



Course Objectives

- **J2EE 1.4/ JavaEE5**/ JavaEE6 Platform API



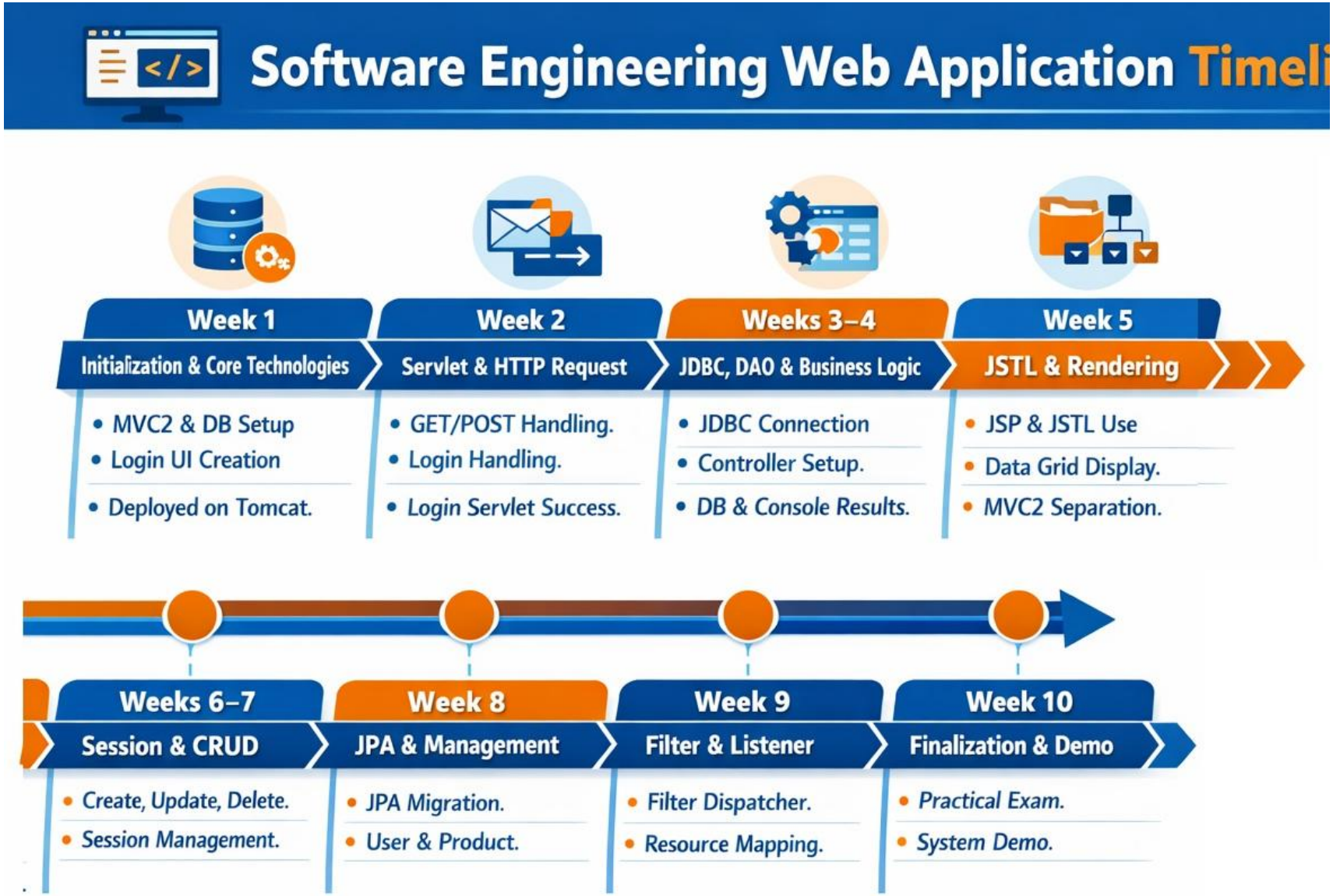
Course Plan

- See course plan on LMS/EduNext
- Agenda

1. The Servlet Model
2. Web Application Interacting with Database
3. Web Application & Web Container
4. Java Server Pages (JSP)
5. Session Management, Session & Listener (self-study)
6. JSP Tag Libraries
7. JPA – Java Persistence API
8. Filters
9. JPA
10. *Practical test*
11. *Project Presentation*

Course Plan

- **See course plan on LMS/EduNext**

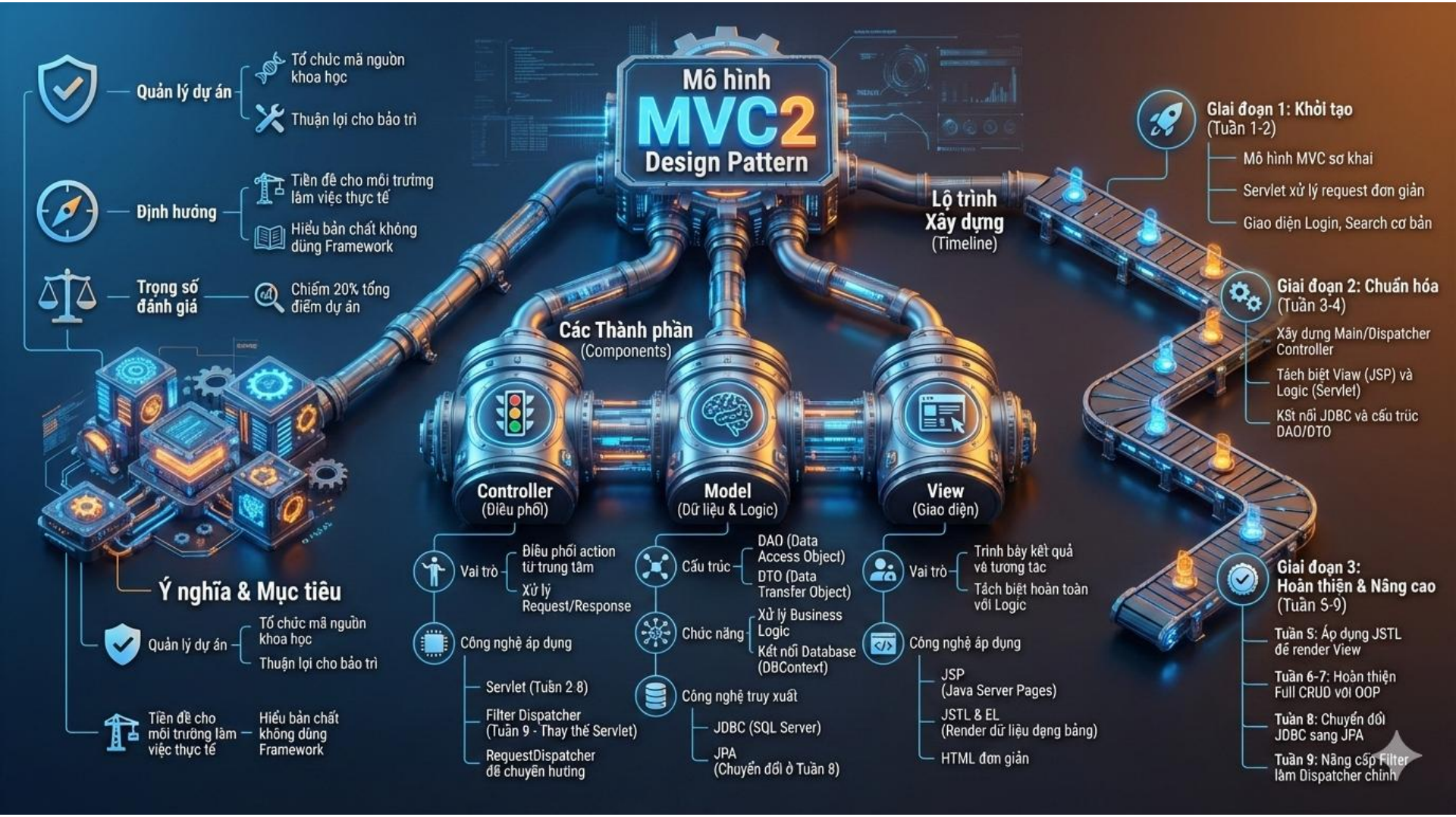


Course Plan

- **See course plan on LMS/EduNext**

Week	Main Topic	Technical Objectives & Tasks	Expected Deliverables
Week 1	Initialization & Core Technologies	- Understand MVC2 model and Web Application architecture. - Create SQL database and initialize project skeleton. - Implement Login UI (interface only).	- Project successfully deployed on Tomcat. - HTML pages: Login, Search, Invalid.
Week 2	Servlet & HTTP Request/Response	- Handle GET/POST methods and URL mapping. - Receive parameters from Login page. - Apply DAO pattern to separate logic.	- Servlet processes successful Login request. - Display received username/password on browser.
Weeks 3–4	JDBC, DAO & Business Logic	- Establish database connection using JDBC. - Construct Main Controller / Dispatcher Controller. - Implement real Login and Search functionality (approximate matching).	- Successful DB connection and data retrieval. - Login and search results displayed in console/log.
Week 5	JSTL & Rendering	- Use JSP as View layer. - Apply JSTL/EL for data presentation.	- Search results displayed as Data Grid (table). '- Ensure MVC2 applying correctly components.
Weeks 6–7	Session Tracking & Full CRUD	- Complete database operations: Create, Update, Delete. - Manage user sessions - Apply OOP thoroughly in DAO and DTO.	- Full CRUD functionality: Add, Update, Delete products based on Primary Key.
Week 8	JPA & Product Management	- Migrate data layer from JDBC to JPA. - Manage User and Product entities.	- Source code for User/Product management using JPA.
Week 9	Filter & Listener	- Replace Servlet Dispatcher with Filter-based Dispatcher. - Enable flexible resource mapping. - Prepare for Practical Exam.	- Project reconstructed with Filter as Main Controller. - Practical exam conducted at the end of Week 9.
Week 10	Finalization & Demo	- Finalize all system functionalities.	- Project evaluation and full system demo.

Course Objectives



Materials/ References

- **Required Textbook**
  - Online Text book: Nicholas S. Williams, 2014, Professional Java® for Web Applications, Wrox Press
  - http://library.books24x7.com/toc.aspx?bookid=62587
- **Required References**
  - http://java.sun.com/docs/books/tutorial/jdbc/
  - https://docs.oracle.com/cd/B14099_19/web.1012/b14017/filters.htm
- **References**
  - http://java.sun.com/
  - Reference Page: https://www.kieutrongkhanh.net/
  - Fan Page: https://www.facebook.com/TrongKhanh.Kieu/

Learning Environments

- JDK 8 (https://www.oracle.com/java/technologies/downloads/)
  - **Recommend**: JDK 8 Update 172 (https://www.oracle.com/java/technologies/javase/javase8u211-later-archive-downloads.html)
- JDK 8 Documentation
- J2EE 1.4/JavaEE7 Core Patterns
- **NetBeans IDE 13 with supporting JavaEE** (https://www.apache.org/dyn/closer.lua/netbeans/netbeans-installers/13/Apache-NetBeans-13-bin-windows-x64.exe)
- **Tomcat 9.0.113** (https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.113/bin/apache-tomcat-9.0.113-windows-x64.zip)
- DBMS: **MS. SQL Server 2019** (http://www.microsoft.com/sqlserver)
- Browser: **Edge** (should have **Internet Explorer simulator)**
- Driver Type 4 for MS. SQL Server: **sqlserver4.jar** (on LMS)
- **GG Meet for supporting (efc-padj-aob)**

Course Rules

How to conduct

Prepare contents of the next session/ topic at home

Following lessons in classrooms (take note yourself to understand using notebooks) and *review previous session in lecture on class every day (penalty marks on workshops)*

At least 02 session for studying social constructivism with deposit for Assignment

Completing chapter assessments in time and Quizzes (via LMS)

Communication

Class, Interchange by FU-HCM LMS, Forum

Discussing actively in your teams and in classrooms

Free to question and answer

Others

Paper notes are required

Off phone

Use laptops under teacher's instruction (No game, no chat in class)

Evaluation Strategy

- Must attend more than 80% of contact hours (if not, not allow to take exam).
- **Evaluating**
  - **02 Progress Test (Q)** 2 x 5 %
  - **02 Workshop (Lab)** 2 x 5 %
  - **01 Assignment (Prj)** 30 %
  - **01 Practical Exam (P)** 30 %
  - **Final Exam (FE)** 20 %
- **Total score**
  - 10% (Q) + 10% (Lab) + 30% (Prj) + 30% (P) + 20% (FE)
- **Pass**
  - Total score ≥ 5 and Final Examination ≥ 4 (of 10)
  - **Every components > 0**
- **Retake** only the Final Exam when not passed

Assignments/Workshops/Quizzes

Assignments	Workshops	Quizzes
• Building project in class (**evaluating face to face**) (50%) • **Some sessions** for social constructivism • **Contribution** in lectures on our class • **Mind Maps** of core flow **& Workshop** (40%) • **Enhance Self Worth - Coursera Certification** (10%)	• **Contribution** in lectures on our class (participation) • **Workshops and Peer Reviews in Weeks**	• Use **Safe Exam Browser** • **guideline with LMS/exam.fpt.edu.vn**

Assignments/Workshops/Quizzes

Assessment Area

Assignment
Opened: Monday, 5 January 2026, 12:00 AM **Due:** Wednesday, 18 March 2026, 11:00 PM

ProgressTest1
Opens: Monday, 23 February 2026, 7:08 AM **Closes:** Monday, 23 February 2026, 7:35 AM

From Topic 1 to 4: Servlet, Web Container, JDBC and JSP

Not available unless: You belong to **SE1921_SP2026**

ProgressTest1to8
Opens: Monday, 16 March 2026, 7:08 AM **Closes:** Monday, 16 March 2026, 8:05 AM

Not available unless: You belong to **SE1921_SP2026**

Workshop 1st Week
Submissions open: Monday, 5 January 2026, 7:50 AM **Submissions close:** Saturday, 10 January 2026, 11:50 PM **Assessments open:** Sunday, 11 January 2026, 7:50 AM **Assessments close:** Wednesday, 14 January 2026, 11:50 PM

Not available unless any of: You belong to **SE1910_SP2026** ... Show more

Workshop 2nd Week
Submissions open: Monday, 12 January 2026, 7:50 AM **Submissions close:** Saturday, 17 January 2026, 11:50 PM **Assessments open:** Sunday, 18 January 2026, 7:50 AM **Assessments close:** Wednesday, 21 January 2026, 11:50 PM

Not available unless any of: You belong to **SE1910_SP2026** ... Show more

Workshop 3rd - 4th Week
Submissions open: Monday, 19 January 2026, 7:50 AM **Submissions close:** Saturday, 31 January 2026, 11:50 PM **Assessments open:** Sunday, 1 February 2026, 7:50 AM **Assessments close:** Wednesday, 4 February 2026, 11:50 PM

Not available unless any of: You belong to **SE1910_SP2026** ... Show more

Workshop 5th Week
Submissions open: Monday, 2 February 2026, 7:50 AM **Submissions close:** Saturday, 7 February 2026, 11:50 PM **Assessments open:** Sunday, 8 February 2026, 7:50 AM **Assessments close:** Wednesday, 11 February 2026, 11:50 PM

Not available unless any of: You belong to **SE1910_SP2026** ... Show more

Assignments/Workshops/Quizzes

Workshop 6th - 7th Week

Submissions open: Monday, 23 February 2026, 7:50 AM **Submissions close:** Saturday, 7 March 2026, 11:50 PM **Assessments open:** Sunday, 8 March 2026, 7:50 AM **Assessments close:** Wednesday, 11 March 2026, 11:50 PM

Not available unless any of: You belong to **SE1910_SP2026** ... Show more

Workshop 8th Week

Submissions open: Monday, 9 March 2026, 7:50 AM **Submissions close:** Saturday, 14 March 2026, 11:50 PM **Assessments open:** Sunday, 15 March 2026, 7:50 AM **Assessments close:** Wednesday, 18 March 2026, 11:50 PM

Not available unless any of: You belong to **SE1910_SP2026** ... Show more

Workshop 9th Week

Submissions open: Monday, 16 March 2026, 7:50 AM **Submissions close:** Saturday, 21 March 2026, 11:50 PM **Assessments open:** Sunday, 22 March 2026, 7:50 AM **Assessments close:** Monday, 23 March 2026, 11:50 PM

Not available unless any of: You belong to **SE1910_SP2026** ... Show more

Future Foundation

Enhance Self-worth

- **Goal**
  - Have certificate similar this course's content
  - Enables to build scalable and accessible enterprise applications
- **Over 20 hours (3 courses)**
  - **https://www.coursera.org/specializations/java-servlet-development-from-basics-to-real-world-projects**
- **Fill your information:** (Deadline **17/01/2026**)
  - https://docs.google.com/spreadsheets/d/1kapXTm_WZokyj2WfEUsxx84Mr5vXAZ94/edit?usp=sharing&ouid=109354261804343881967&rtpof=true&sd=true
- **Benefits (Deadline Submission 14/03/2026)**
  - Deposit 10% project, assignments, … when students take certification

Future Foundation

Enhance Self-worth

Java Servlet Development: From Basics to Real-World Projects Specialization

Java Servlets Tutorial

Course 1 • 17 hours

Java Servlets Case Study - Email Marketing Tool

Course 2 • 5 hours

Java Servlets Case Study - Online Admission Process

Course 3 • 7 hours

How to study

- This course is **complex knowledge** (**however**, it's **attractive and exciting**), so you need to keep tight grip on it
  - **Read**
    - On the books to get the general concept
    - Reference, study, collection from internet, your classmates, forum …
  - **Attend lectures**
    - Listens, understand, then **make your own notes (requir**
    - Give your explanation about some topic in lectures
    - Ask questions
    - Give some examples that are not existed in your book
    - Practice all the exercises, demo to make your sense
  - **After classes**
    - Discuss your classmate in directly, on forum
    - Do the lab, assignments to submit via CMS, and do more exercises
    - Build your teams in yourselves to support together in studying

→ Make question to deposit marks for components in progress

→ 99% attendance participant will be deposited 01 mark in Assignment

How to exam/test

- This course is **required following rules**, so you **need to focus and practice** your exercises and homework in try your best everyday (**meet deadline and submit on LMS**)
  - **Progress Tests (if students violate, they take 0 marks)**
    - **No** books and **No** conversations, must be used SEB
  - **Workshop/Assignment (if students violate, they take 0 marks)**
    - **Obey requirements and lecturers recommended in his lecture**
    - **Not** copy (copy code, contents, style)
    - Submission of all source code **does not delete anything**
  - **Practical Exam (if students violate, they take 0 marks)**
    - Based on regular of Examination Dept.
    - Some rules and recommended thing in LMS

Academic policy

- Cheating, plagiarism and breach of copyright are serious offenses under this Policy.
  - **Cheating**
    - Cheating during a test, making a project, or an exam is construed as talking, peeking at another student's paper, or any other clandestine method of transmitting information.
    - Cheating during in making lab and assignment as copy source code, copy style, same meaning in progress, …
    - **To verify, student's code in project can be deleted, then he/she must be retyped to make the program running correctly. Or/And make new required functions (Otherwise, he/she takes 0 marks for his/her project/quiz)**
  - **Plagiarism**
    - Plagiarism is using the work of others without citing it; that is, holding the work of others out as your own work.
    - **To verify, student must be described functionality dataflow and how it work (Otherwise, he/she takes 0 marks for his/her project)**
  - **Breach of Copyright**
    - If you photocopy a textbook without the copyright holder's permission, you violate copyright law.

Enjoy the Course

Be enthusiastic about the material because it is interesting, useful and an important part of your training as a software engineer.

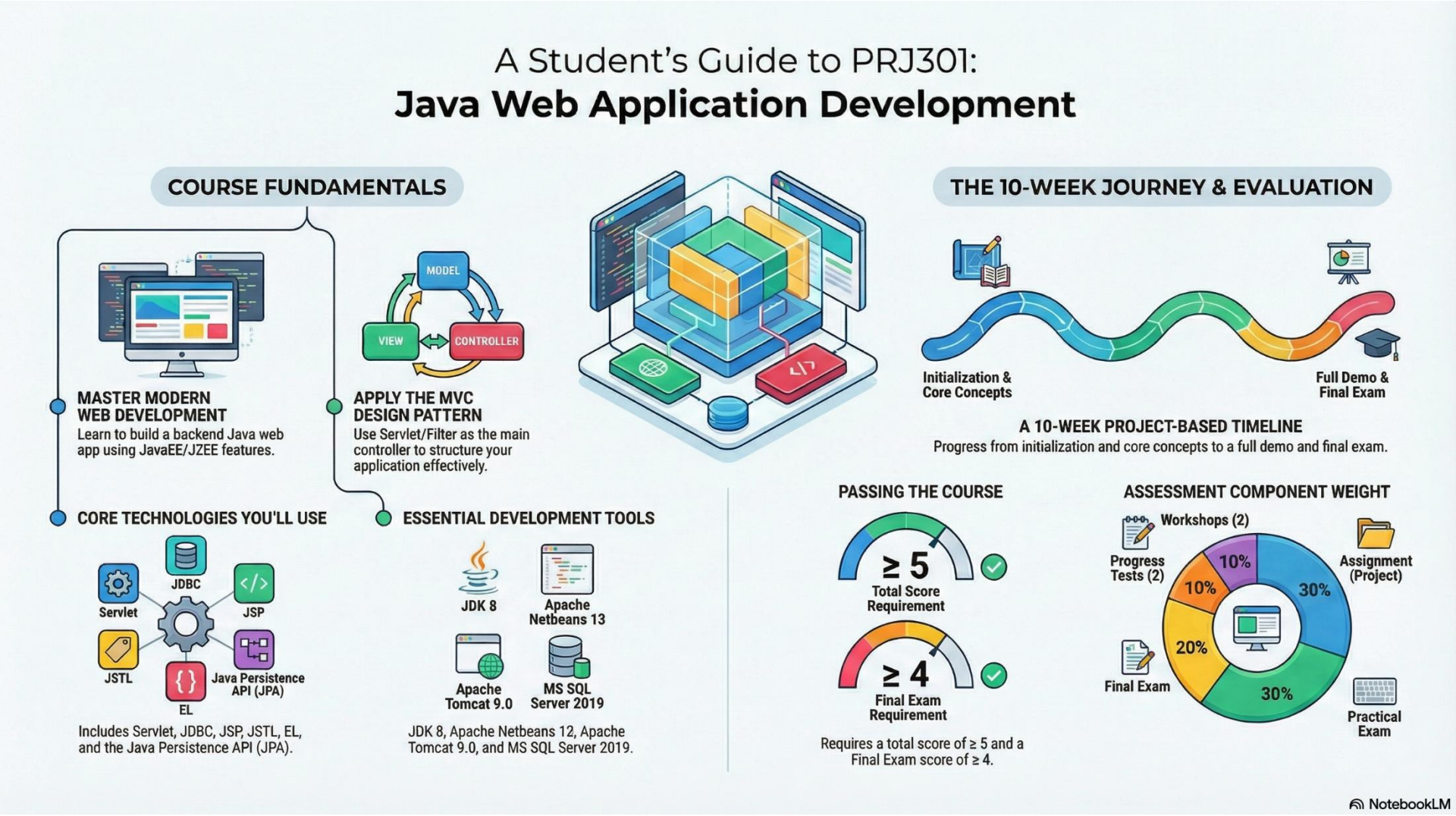
Our job is to help you learn and enjoy the experience.

We will do our best, but we need your help.

So, let’s all have fun together with
Web-Based Java Applications!!!

Fpt University

Q & A

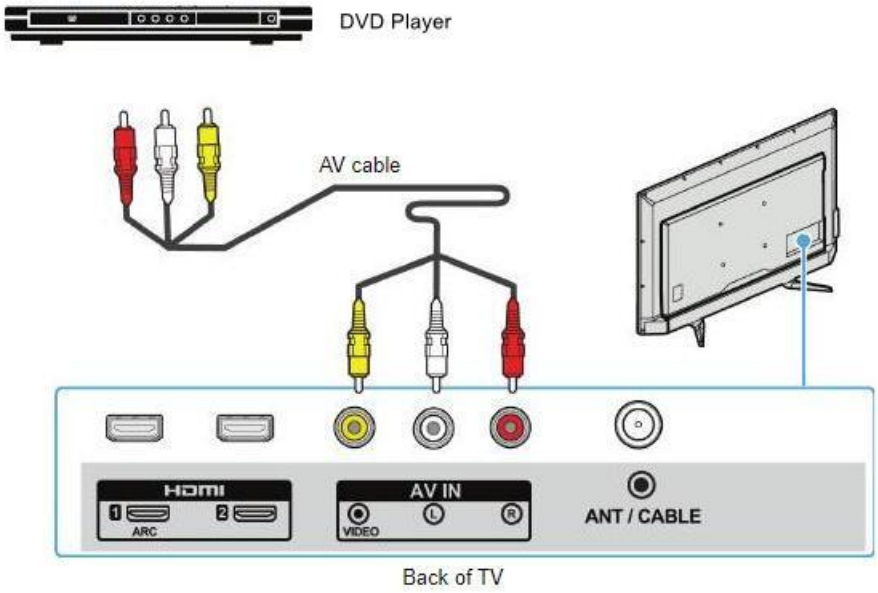
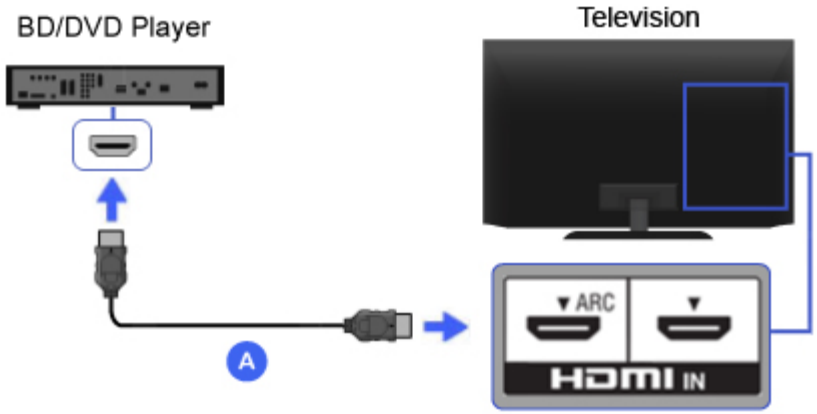


MVC Design Pattern

Remote Controller

MVC Design Pattern

Player

MVC Design Pattern

Connection